

Unit

3

Lesson

2



Beyond Imagination.

Find Your Way.

Time Frame: [90 min]

Lesson objectives:

By the end of this lesson students will be able to:

- Know what coding is and how it helps characters move and solve problems.
- Use mBlock to create and play a simple maze game.
- Use motion blocks to move characters forward, backward, left, and right.
- Understand X and Y directions to guide the character through the maze.
- Use costumes to change how the character direction.
- Add clicking events and sensing blocks to make the game more fun.
- Put coding steps in the right order (sequencing).
- Use algorithms to give the character a list of clear steps.

Challenge questions of the lesson

1. Why is it important for your code to be in the right order for Khaled to move correctly? What if you put the steps in the wrong order?
2. Imagine Khaled is supposed to move 'right' but instead he moves 'down'. What could be a simple mistake in your code that would make him do that?

SEL Relation

- **Being Kind and Fair:** We think about how to make games and places fun and easy for everyone, even those who move differently. This helps us be kind and fair to all our friends.
- **Solving Problems:** When our code doesn't work right, we learn to keep trying and find ways to fix it. This helps us become better problem solvers in our games and in life.
- **Working with Others:** When we work on projects, we learn to share our ideas and help our friends. This makes us good teammates and helps us get along.

Engage



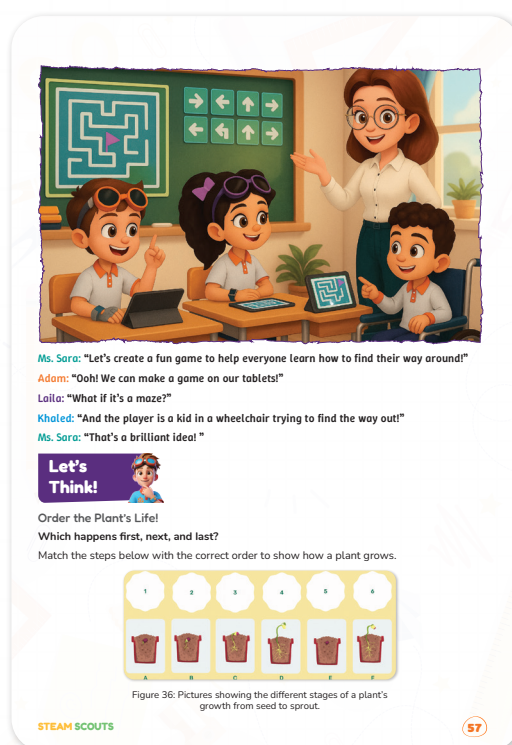
Teacher Role:

- **Storyteller & Scene Setter:** Begin by engagingly reading or narrating the "Miss Sarah" dialogue, setting the stage for the new project.
- **Facilitator of Ideas:** Encourage students to actively listen to the dialogue and then brainstorm their own ideas about how they can contribute to making a game that helps everyone.
- **Questioner:** Ask open-ended questions to probe students' understanding of the story and encourage them to connect with the ideas of inclusivity and problem-solving through design.
- **Enthusiasm Builder:** Show genuine excitement for the "brilliant idea" from the students in the story, inspiring the class to embrace the challenge.

Expected From Students:

- **Active Listening:** Students will listen attentively to the "Intro Story" dialogue.
- **Idea Generation:** Students will think about and potentially share their initial ideas for how to create a helpful game, especially one for people with different needs.
- **Engagement with Dialogue:** Students will react to the ideas presented by Miss Sarah, Adam, Laila, and Khaled.
- **Connecting to Real-World Needs:** Students will begin to understand the concept of designing for accessibility, particularly related to movement and navigation.

Parts of the books included.



Story



Hook the Class with Imagination:

“Imagine you’re at a brand-new playground... but there are no signs or sounds to help you know where to go. What if someone in a wheelchair was trying to find the slide — how could we help them?”

Introduce the Scenario:

“In today’s story, our friends had an awesome idea — they’re going to design a maze game where the player is a kid in a wheelchair trying to find their way out. This is our challenge too!”

Connect to the Lesson Goal:

“We’ll use tablets or paper to design a maze that is fun and fair. It should help everyone — including kids who use wheelchairs — find their way. Are you ready to become game designers?”

Possible Strategies for Implementation

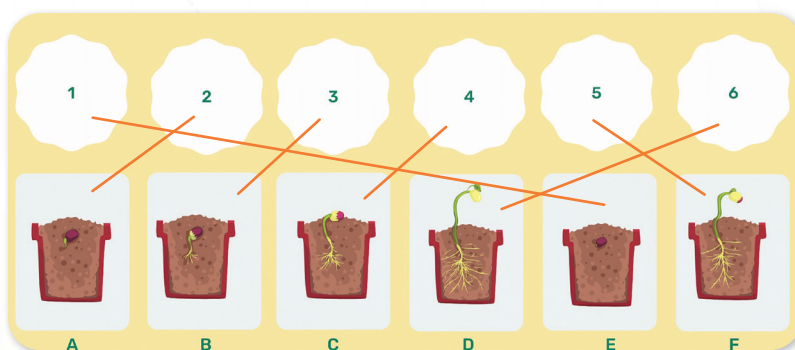
- Story-Driven Inquiry (Engage Strategy):
 - Read the dialogue dramatically with voices.
 - Pause after Khaled’s line and ask: “What kinds of challenges would someone in a wheelchair face in a maze?”
 - Let students brainstorm what would make a maze inclusive.
- Think-Pair-Share:
 - “What features would you put in your game to help everyone move safely?”
 - Let students pair up and share their answers before whole-class discussion.

Let’s Think!



Objective: Students will strengthen their understanding of sequencing by identifying and ordering the correct steps in a plant’s life cycle from seed to sprout.

Answer:



Activity instructions

- Introduce the Concept of Sequencing

Begin with a short class discussion:

- “What does it mean to put things in the right order?”
- “Why is order important when we talk about how things grow?”
- Guide the Visual Observation

Ask students to look carefully at each picture labeled A–F.

Encourage them to describe what is happening in each one (e.g., “This one has just a seed,” “This one has a stem growing.”)

- Explain the Task
 - Tell students to write the correct number (1 for first, 6 for last) above each picture to show the correct order of how a plant grows.
 - Remind them: “Start with the seed. What happens next?”
- Encourage Thinking Aloud

While students work, walk around and ask them to explain their choices using sequencing words:

- “First... then... next... after that... finally...”
- Wrap-Up Discussion

After everyone finishes, go through the correct order together as a class and invite volunteers to explain why they chose that order.

Explorer's Toolkit



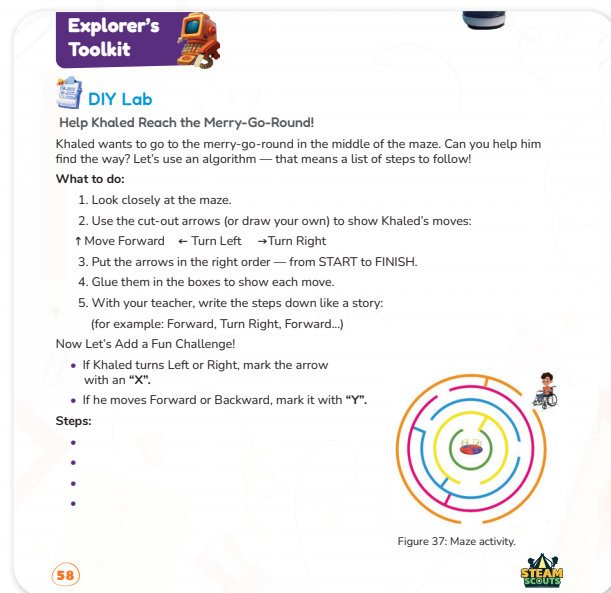
Teacher Role:

- **Instruction Clarifier:** Clearly explain the task of helping Khaled navigate the maze using an algorithm, which is a list of steps to follow.
- **Modeling/Demonstration:** Show an example of how to use the arrows to represent moves and turns on a small part of the maze or a similar simple path.
- **Algorithm Guide:** Assist students in understanding how to translate movements into sequential steps (an algorithm). Guide them in writing down the steps in a clear, ordered list (e.g., “Forward, Turn Right, Forward...”).
- **Troubleshooting Support:** Help students when they get stuck, prompting them to re-evaluate their current position and the next necessary move.
- **Assessment Monitor:** Circulate and observe students' understanding of sequencing and their ability to apply directional commands.
- **Extension Prompt:** Introduce the “Add a Fun Challenge!” part by explaining how to mark arrows for turns (“X”) and forward/backward moves (“Y”).

Expected From Students:

- **Maze Analysis:** Students will carefully observe the maze to plan Khaled's path.
- **Algorithm Creation:** Students will use cut-out arrows or draw their own to show Khaled's moves in the correct order.
- **Sequencing:** Students will put the arrows in the right order from START to FINISH.
- **Hands-on Participation:** Students will actively manipulate the arrows (cut-outs or drawn) and glue them in the boxes to show each move.
- **Written Algorithm:** Students will work with the teacher to write down the steps like a story (e.g., Forward, Turn Right, Forward...).
- **Challenge Application:** Students will mark their arrows with an "X" for turns and a "Y" for forward/backward moves as an extra challenge.
- **Problem-Solving:** Students will attempt to find the correct path and correct any errors in their sequence.

Parts of the books included.



DIY Lab

Objective: Students will practice sequencing and basic algorithmic thinking by creating and following a step-by-step path using directional arrows to help Khaled reach the merry-go-round. This activity supports logic development, direction recognition, and understanding how instructions guide movement — a foundation for early coding skills.

Answer:

Steps to Reach the Merry-Go-Round:

1. Move Forward →
2. Turn Left →
3. Move Forward →
4. Turn Right →

5. Move Forward →
6. Turn Right →
7. Move Forward

With Labels (for the challenge part):

- Step 1: ↑ → Y
- Step 2: ↶ → X
- Step 3: ↑ → Y
- Step 4: ↷ → X
- Step 5: ↑ → Y
- Step 6: ↷ → X
- Step 7: ↑ → Y

Activity Instructions

1. Introduce the Challenge:

Say: “Khaled wants to go to the merry-go-round, but there’s a maze in the way! Let’s help him get there using steps — like a computer program!”

2. Demonstrate the Moves:

- Show the meaning of each arrow:
 - ↑ = Move Forward
 - ↶ = Turn Left
 - ↷ = Turn Right

3. Distribute Materials:

- Printed maze (from the book)
- Arrow cutouts or blank paper for students to draw their own
- Scissors and glue
- Pencils for writing steps

4. Guide Students Through the Maze:

- Let students trace Khaled’s possible route using a finger or pencil.
- Encourage them to decide the correct direction at each point.

5. Create the Step Sequence:

- Instruct students to either cut and glue arrows in order, or draw them in the boxes labeled “Steps.”
- Emphasize correct sequencing — what happens first, next, then last?

6. Label the Moves:

- For each move, ask students to mark:

- "X" for a Turn (Left or Right)
- "Y" for Straight moves (Forward or Backward)

7. Write it Like a Story (Optional Extension):

- Students work with a partner or teacher to write their moves as a sentence:

"Forward, Turn Left, Forward, Turn Right..."



Teacher Role:

- **Concept Introducer:** Introduce the video as a way to learn more about "sequencing" and how it helps us tell computers what to do.
- **Purpose Setter:** Clearly explain why students are watching the video and what they should pay attention to.
- **Discussion Facilitator:** Guide a discussion after the video to check for understanding of "sequencing" and "algorithm".
- **Vocabulary Clarifier:** Use the "Wrap-Up Time!" definitions to reinforce key terms like "sequencing," "algorithm," "X-axis," and "Y-axis".
- **Real-World Connector:** Help students connect the abstract concepts of sequencing and algorithms to everyday actions and the maze activity they just completed.

Expected From Students:

- **Active Viewing:** Students will watch the video with a purpose, looking for how steps are put in order.
- **Concept Grasping:** Students will develop an understanding of what "sequencing" and "algorithm" mean.
- **Observational Learning:** Students will observe how a character moves on a grid and how that relates to X and Y axes.
- **Discussion Participation:** Students will engage in discussions, sharing their observations and understanding of the concepts.

Parts of the books included.

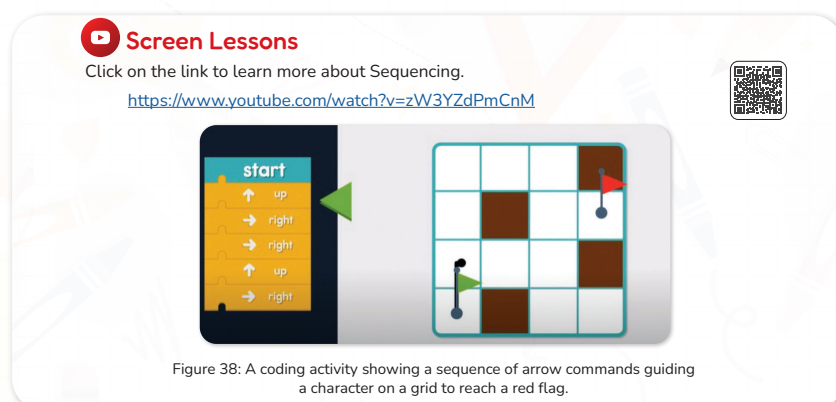


Figure 38: A coding activity showing a sequence of arrow commands guiding a character on a grid to reach a red flag.

Screen Lessons

Objective: Students will understand the concept of sequencing as putting steps in the correct order by watching real-life examples. This will help them apply sequencing skills in problem-solving tasks such as giving directions, solving mazes, or building projects step-by-step. The video supports logical thinking and prepares students to use “first, next, last” in both verbal and written instructions.

Video link: <https://www.youtube.com/watch?v=zW3YZdPmCnM>



Possible Strategies for implementation:

1. Pre-Watching: Set a purpose.

- Strategy: "Predict the Path"
 - How to Implement: Display the image of the coding blocks and grid from "Screen Lessons". Ask students: "Look at the 'start' block and the arrows below it. What do you think those arrows will make the character do? What path do you think it will take on the grid?" Introduce the term "sequencing" as putting steps in the right order. Tell them they will watch a video to learn more about how "sequencing" helps characters move.
 - Benefit: Engages students by having them predict outcomes, connecting to their previous "Do It" maze activity, and introduces the key vocabulary.

2. During Watching: Active engagement.

- Strategy: "Word Watch"
 - How to Implement: Tell students to listen specifically for the words "sequencing" and "algorithm" in the video. When they hear it, they can give a quiet thumbs-up.
 - Benefit: Keeps students focused on identifying key vocabulary and reinforces the importance of these terms.

3. Post-Watching: Discussion and reflection.

- Strategy 1: "Define It Together"
 - How to Implement: After the video, ask students: "What does 'sequencing' mean?" and "What is an 'algorithm'?" Guide them to use the definitions provided in the "Wrap-Up Time!" box: "Sequencing means putting steps in the right order. Algorithm is a list of steps to solve a problem or do something."
 - Benefit: Solidifies understanding of core vocabulary directly from the provided lesson materials.
- Strategy 2: "Everyday Algorithms"
 - How to Implement: Ask students to brainstorm simple "algorithms" they use every day, like getting ready for school (brush teeth, get dressed, eat breakfast) or making a sandwich (get bread, put on peanut butter, put on jelly, close bread). Discuss why the order matters.
 - Benefit: Makes the concept of algorithms relatable and reinforces that they are already using these sequential thinking skills.

Elaborate



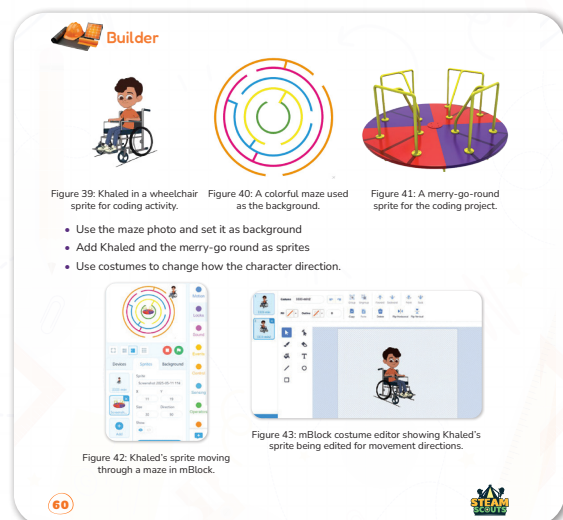
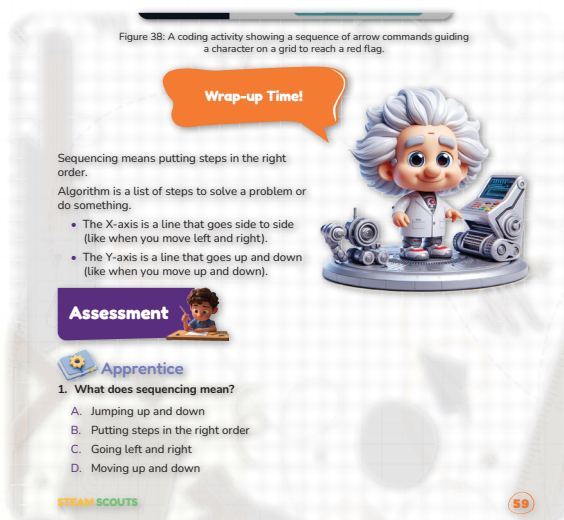
Teacher Role:

- Facilitate assessment of students' understanding of key vocabulary such as sequencing, algorithm, and X-axis using multiple-choice questions.
- Provide constructive feedback and clarify any misconceptions based on students' answers.
- Guide students in using the Scratch environment by helping them:
 - Set up their workspace
 - Import backgrounds and sprites
 - Use costume changes to show character direction
- Support students in solving technical and conceptual challenges during coding.
- Encourage creativity and digital literacy by promoting exploration within Scratch.
- Observe and assess students' ability to apply sequencing and algorithms to simulate real-world maze navigation digitally.

Expected From Students:

- Demonstrate understanding of key concepts by correctly answering multiple-choice questions on sequencing, algorithm, and X-axis.
- Apply their knowledge of sequencing by planning a clear path through a maze.
- Translate a physical or visual path into a Scratch project using code blocks and sprites.
- Import and manipulate sprites and backgrounds in Scratch to match the maze design.
- Use costume changes and direction controls to animate character movement.
- Debug and refine their project to ensure the sprite follows the correct sequence.
- Present or explain how their code represents a step-by-step algorithm.

Parts of the books included.





Apprentice

Objective: Students will demonstrate their understanding of foundational coding vocabulary—sequencing, algorithm, and X-axis—by correctly identifying definitions and directional concepts through a multiple-choice assessment. This activity reinforces computational thinking skills and prepares students for digital programming tasks like creating movement-based projects in Scratch.

Possible answers:

1. What does sequencing mean?

- A. Jumping up and down
- B. Putting steps in the right order
- C. Going left and right
- D. Moving up and down

2. What is an algorithm?

- A. A type of robot
- B. A list of steps to solve a problem or do something
- C. A type of dance
- D. A list of favorite foods

3. Which direction does the X-axis go?

- A. Up and down
- B. Side to side (left and right)
- C. Forward and backward
- D. Around in a circle

Possible Strategies for Implementation:

- Preview Vocabulary: Before beginning, review the terms sequencing, algorithm, and X-axis with examples and visuals (e.g., arrows, steps, simple robot path).
- Think-Pair-Share: Have students discuss each question in pairs before selecting their answers. This encourages peer explanation and collaborative reasoning.
- Interactive Anchor Chart: Create a class chart with student-generated definitions and illustrations for each term that they can refer to during the activity.
- Mini-Movements: Use kinesthetic learning by asking students to physically move side-to-side to represent the X-axis and act out “sequencing” using step cards.
- Immediate Feedback: Review answers together after completion, discussing why each correct answer makes sense and clarifying misconceptions as needed.
- Bridge to Digital Coding: Relate answers to what they will do in m.block e.g., “We’ll use X-axis knowledge when moving sprites left or right.”



Objective: Students will apply foundational programming concepts by designing and coding a maze navigation game in mBlock, where a character sprite (Khaled) moves toward a goal (merry-go-round). They will use sequencing, sprite manipulation, and direction changes to simulate navigation using visual code blocks.

Software link: <https://mblock.cc/pages/downloads>

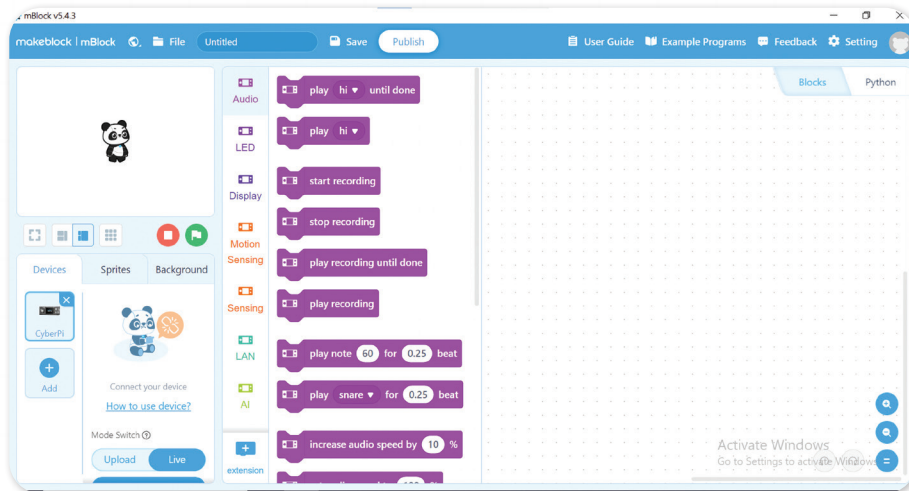


Resources link: <https://drive.google.com/file/d/1m38Hna5UYhzajzwgMWi9JlxDScm-ou-L/view>

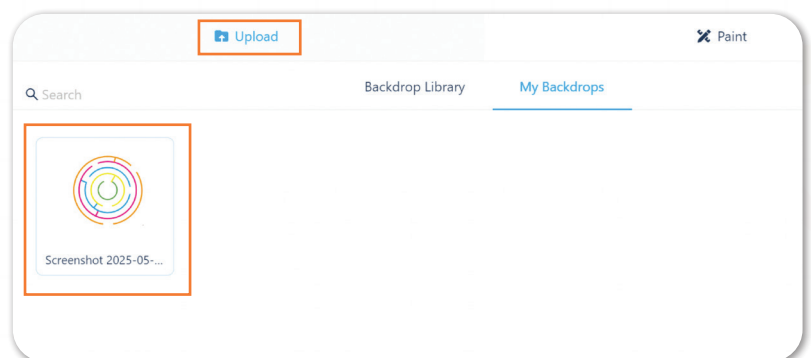
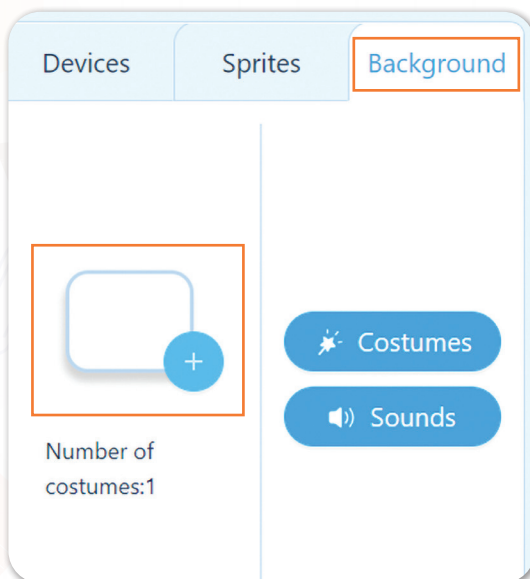


Activity instructions:

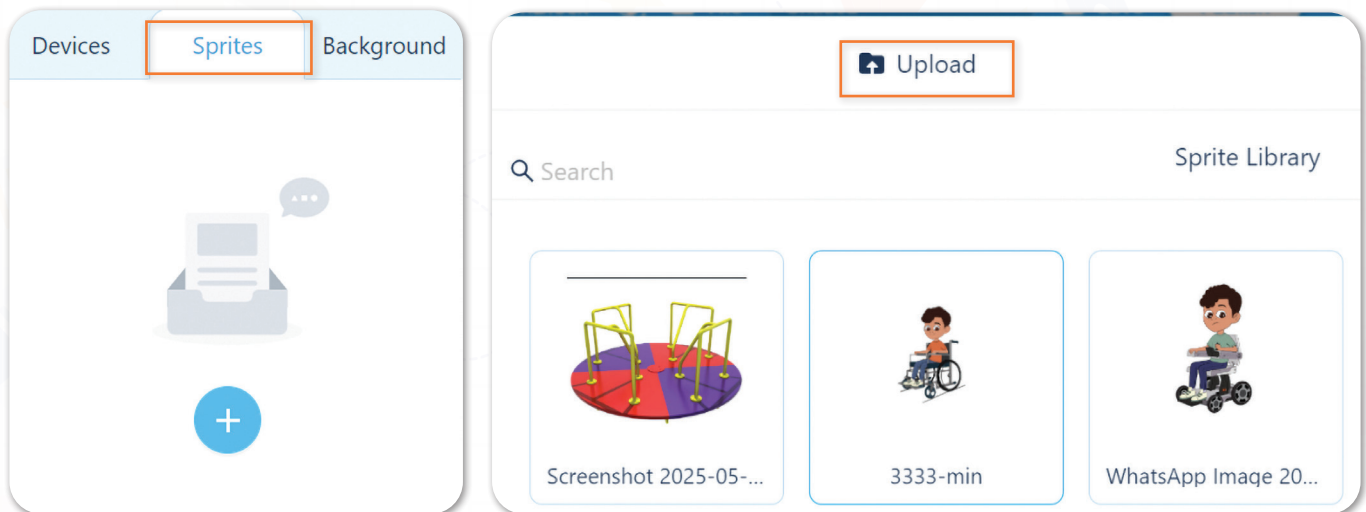
- Open mBlock on your device.



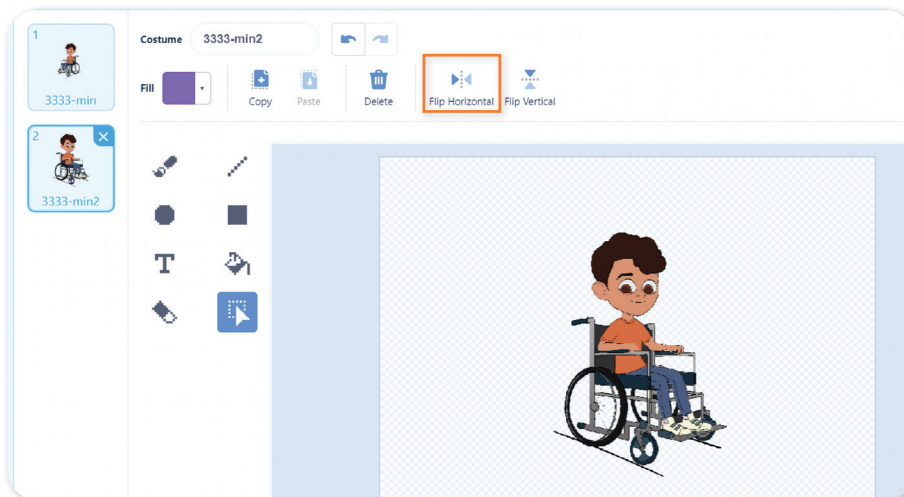
- Set the Background:
 - Upload the provided maze image and set it as your game's background.



- Add Sprites:
 - Insert Khaled as the main character sprite.
 - Add a merry-go-round sprite and place it at the maze's center or endpoint.



- Use Costumes for Direction:
 - Use Khaled's costume changes to show direction



Evaluate



Teacher Role:

- **Coding Facilitator:** Guide students as they arrange code blocks to make Khaled move right, left, up, and down.
- **Logical Thinker:** Help students understand the sequence of code blocks required for each movement and turn, connecting it to the concepts of algorithms and sequencing.
- **Troubleshooting Assistant:** Provide support when students encounter errors in their code, prompting them to debug and re-evaluate their block arrangements.
- **Encourager of Experimentation:** Foster a safe environment for students to try

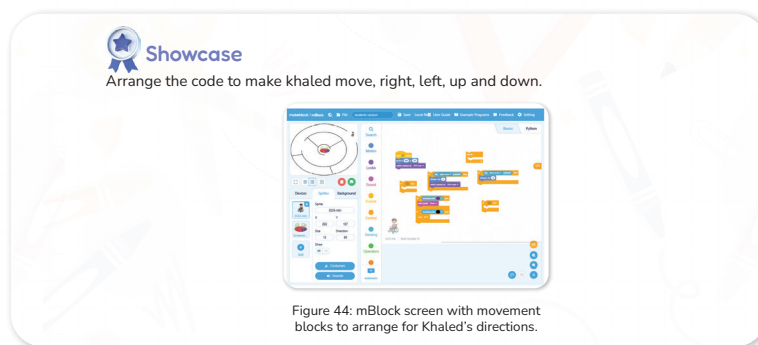
different code combinations and observe the results.

- **Evaluator:** Use the provided rubric to assess students' ability to correctly arrange code blocks and demonstrate character movement.
- **Connection Maker:** Remind students how the digital movement relates to the physical maze activity they completed earlier.

Expected From Students:

- **Code Arrangement:** Students will arrange pre-provided or newly selected code blocks in Scratch to control Khaled's movement.
- **Directional Coding:** Students will demonstrate understanding of how to make Khaled move right, left, up, and down using appropriate code blocks.
- **Application of Sequencing:** Students will apply their knowledge of sequencing to ensure the code blocks are in the correct order for desired movement outcomes.
- **Problem-Solving & Debugging:** Students will identify and attempt to fix errors in their code to achieve the intended movement.
- **Demonstration of Understanding:** Students will showcase their coded Khaled moving successfully in different directions.

Parts of the books included



Objective: Students will explore directional movement by arranging block-based code in mBlock to move a character (Khaled) in the directions: right, left, up, and down. This activity strengthens sequencing, direction recognition, and coding fluency using visual programming.

Activity instructions:

Preparation Before Class

- Open the Scratch or mBlock project with the scrambled blocks (as shown in the image).
- Ensure each student or pair has access to the project on their screen.

During the Lesson

1. Introduce the Goal:

Explain that Khaled wants to move through a path using arrow keys. Students must

arrange the blocks so:

- Arrow keys move him left, right, up, and down.
- He changes costume to show direction.
- He reacts to different colors (purple = cheer, black = stop).

2. Guide the Process:

- Ask: “What do you think this block does?” for each type (motion, looks, sensing, control).
- Discuss X vs. Y axis movement.
- Help students test their code and debug if Khaled moves incorrectly.

3. Encourage Peer Talk:

- Allow students to discuss where blocks should go before dragging.
- Use questions like “What happens when we press this key?” or “What should Khaled do when he touches black?”

4. Support Independence:

- Offer gentle hints instead of direct answers. For example:

“Which block would help Khaled go up?” or

“What do you notice when he touches purple?”

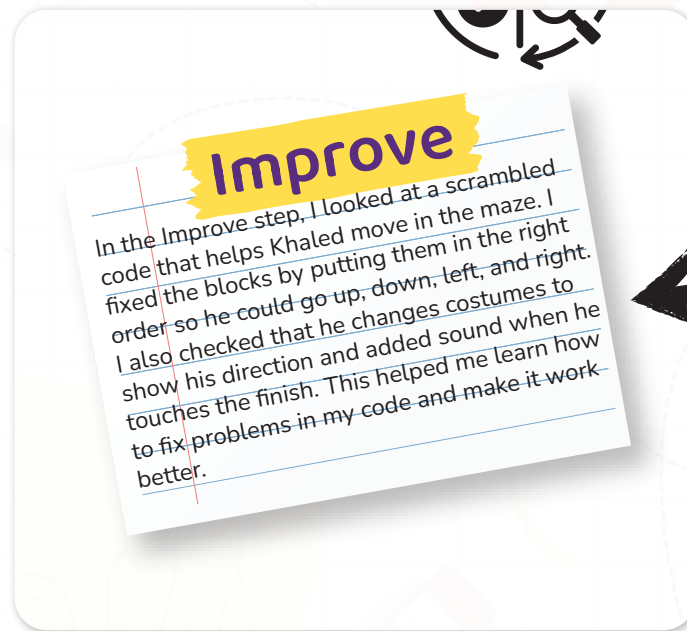
Closure

- Have students demo their working code.
- Discuss how sequencing matters in coding.
- Reflect: “What did Khaled do when you pressed the keys?” “What happens when order is wrong?”

Rubric for Showcase Evaluation:

Criteria	1 Point (Developing)	2 Points (Approaching)	3 Points (Proficient)	4 Points (Exemplary)
Code Arrangement	Code blocks are disorganized or incorrectly linked; character does not move.	Some code blocks are linked, but movement is inconsistent or incorrect.	Code blocks are mostly arranged correctly, allowing for some intended movement.	Code blocks are perfectly arranged and organized, resulting in smooth and accurate movement.
Directional Control	Khaled does not move in the specified directions (right, left, up, down).	Khaled moves in 1-2 correct directions, but others are incorrect or missing.	Khaled moves correctly in 3 specified directions (right, left, up, or down).	Khaled moves correctly in all four specified directions (right, left, up, and down).
Application of Sequencing	Code order is illogical; character moves randomly or not at all.	Sequencing is attempted but often leads to unintended movement.	Sequencing of code blocks allows for mostly predictable movement, with minor glitches.	Sequencing of code blocks is flawless, leading to precisely executed movements as intended.
Problem-Solving & Debugging	Unable to identify or correct errors, requires constant teacher intervention.	Identifies some errors but struggles to correct them independently.	Identifies and corrects simple coding errors with minimal guidance.	Independently identifies and effectively corrects complex coding errors.
Understanding of Concepts	Cannot explain how the code makes Khaled move or what an algorithm is.	Can vaguely explain some code functions but struggles with "sequencing" or "algorithm."	Can clearly explain how arranging the code blocks makes Khaled move and articulate the meaning of "sequencing" and "algorithm."	Clearly articulates how the code implements an algorithm through sequencing to control movement, demonstrating deep understanding.

It's time to move to the improve part and add this



Now I Can

Objective:

Students will reflect on their understanding of coding by identifying key programming concepts such as motion, direction (X and Y axes), costume changes, and event/sensing blocks used to create a maze game in mBlock. This activity promotes metacognition, reinforces basic coding logic, and helps students evaluate their progress in designing functional, interactive games.

Activity Instructions:

- Review Key Concepts
briefly review with students what they've learned about motion blocks, directions (X and Y), costumes, and events in mBlock.
- Reflect on Learning
Ask students to read each statement and place a checkmark next to the skills they now feel confident about.
- Encourage Examples
Invite students to explain one or two checked items out loud or in their journals. For example:
"I used the change x by 5 block to move Khaled to the right."
- Wrap-Up Discussion
Facilitate a class conversation about how coding helped solve problems and how they could improve their game next time.

Relation to Standards:

1. NGSS (Next Generation Science Standards)

- **K-2-ETS1-1 (Engineering Design)**

Standard: Ask questions, make observations, and gather information about a situation people want to change to define a simple problem.

In this lesson:

- **K-2-ETS1-2 (Engineering Design)**

Standard: Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function.

In this lesson:

- **K-2-ETS1-3 (Engineering Design)**

Standard: Analyze data from tests of two objects designed to solve the same problem to compare strengths and weaknesses.

- **Computer Science Standards**

Computational Thinking – Algorithms and Programming

Standard: Students design and implement simple algorithms using visual code blocks to achieve specific goals.

- **Control Flow**

Standard: Students learn that programs execute commands from top to bottom and that specific blocks control actions.

- **Mathematics Connections**

Coordinate Plane – Early Concept

Standard: Recognize and describe position and direction using basic spatial terms (up, down, left, right).

2. ISTE Standards for Students

- **Empowered Learner (1a, 1c)**

Students set learning goals, use digital tools to explore outcomes, and reflect on results.

In this lesson: Students plan their character's motion, test their algorithm, and explain how sequence affects movement.

- **Innovative Designer (4a, 4b, 4c)**

Students apply a design process to identify problems, test solutions, and reflect on improvements.

In this lesson: Students follow the design–test–improve cycle by adjusting code blocks until their program performs correctly.

Final Scene

Objective:

Students will celebrate the successful completion of their inclusive playground project by reflecting on their teamwork, creativity, and problem-solving using the Engineering Design Process. This activity encourages pride, inclusivity, and a sense of accomplishment while reinforcing key takeaways from the entire unit.

How the Teacher Should Do It:

1. Create a Celebration Atmosphere:

- Set up a display area with students' models, drawings, and digital designs.
- Prepare certificates, stickers, or small awards (optional).



Ms. Sarah: "Look what we've built together! What do you think?"
 Adam: "It's amazing! Everyone can play!"
 Laila: "We made a playground that's fair for everyone!"
 Khalid: "This is the best playground ever!"
 Ms. Sarah: "What did we learn by working together?"
 Adam: "We can solve problems with the Engineering Design Process!"
 Laila: "And everyone deserves to have fun!"

2. Read the Story Aloud:

- Use the script in the image as a class performance or read-aloud moment.
- Let different students read the voices of Miss Sarah, Adam, Laila, and Omar.
- Emphasize the inclusive and collaborative message.

3. Facilitate a Reflection Circle:

- Ask: "What part of the project made you proud?"
- "How did we make sure everyone could play?"
- "What was the hardest thing you solved?"

4. Take Photos or Record the Moment:

- Capture the showcase to share with families or school displays.

5. Celebrate the Process, Not Just the Product:

- Highlight how students followed the Engineering Design Process from Ask to Improve.
- Reinforce the value of empathy, teamwork, and design thinking.



Mission Innovate

Final Mission – Design an Accessible Public Space

Overview:

This is a culminating, open-ended project that allows students to apply all the skills they've gained throughout the unit. Students will take full ownership of the process—from identifying a real-world problem to creating a model and presenting their solution.

Project duration : 2 weeks

- Teacher's Role:

○ **Facilitator of Inquiry:**

Introduce the challenge by reading the mission prompt aloud. Help students understand that their task is to redesign a public space to make it inclusive for everyone, especially those with disabilities.

○ **Encourager of Empathy and Creativity:**

Prompt students to think about the real challenges people face in everyday environments. Encourage them to brainstorm thoughtful solutions.

○ **Supporter of Student-Led Work:**

Step back and let students take charge. Offer support only when needed—remind them to apply what they’ve learned about ramps, circuits, sequencing, space planning, and inclusive design.

○ **Guide for Reflection:**

Encourage students to use the reflection questions on the worksheet and refer to the rubric during the design process.

○ **Assessment Guide:**

Use the rubric provided to assess each group’s final model based on science, technology, engineering, art, math, and communication.

Implementation Tips:

- Let students work in pairs or small groups to foster collaboration.
- Have all previous materials and tools (e.g., craft supplies, reused items, LED or buzzer kits, cardboard) available.
- Use “studio time” for students to build at their own pace.
- Allocate enough time across multiple sessions.
- Celebrate creativity and progress more than perfection.